

MODULAR RETAIL SYSTEM WITH ADAPTIVE ARCHITECTURE, AI-DRIVEN PRICING, AND INTEGRATED CONSUMABLES MANAGEMENT

ABSTRACT

An autonomous retail ecosystem is disclosed that includes a reconfigurable cabinet chassis and modular shelving supporting interchangeable storage and processing modules. In some embodiments, the ecosystem includes a hybrid thermal architecture in which items are stored in a chilled zone and selectively processed in an on-demand heating module, such as an induction heating chamber configured to heat a susceptor-bearing package according to a product-specific profile associated with an identifier. The ecosystem may include negative-pressure ventilation to evacuate steam and odors during heating, optional sanitation purge handling for un-retrieved heated items, and inventory and access control using a distributed sensor network including RFID/NFC readers, shelf or edge antenna arrays, cameras, door status sensors, and electronic locks. An edge computing controller may execute transaction workflows in online and offline modes, buffer audit logs, and coordinate pricing, planograms, and space-yield optimization across a master cabinet and one or more slave cabinets in a modular deployment.

FIELD

The present disclosure relates to automated and unattended retail systems. More particularly, the disclosure relates to modular retail cabinets and kiosks having integrated sensing, secure access control, thermal processing for temperature-sensitive goods, and computerized workflows for inventory, pricing, and consumer transactions.

BACKGROUND

Automated vending and unattended retail systems are increasingly used to distribute consumer goods in locations where staffing is impractical or inefficient. Conventional systems may suffer from duplicated components, limited scalability, and inadequate control of temperature-sensitive goods. For example, heating prepared foods within the same enclosure used for cold storage can introduce moisture, odors, and condensation that degrade stored inventory and create sanitation concerns. Further, many systems lack robust mechanisms to confirm that a user removed only authorized items, to accurately associate expiry or lot information with inventory, or to continue operating reliably when a network connection is unavailable. There remains a need for a modular autonomous retail architecture that supports hybrid thermal zoning, stronger identity and inventory control, and improved utilization of cabinet volume and shelf space while preserving a practical cost structure.

BRIEF DESCRIPTION OF THE DRAWINGS

The following figures are conceptual and illustrative. The system may be implemented without requiring any specific layout or physical arrangement.

FIG. 1 is a front view of an example autonomous retail cabinet illustrating a reconfigurable chassis (“transformer base”) with vertically repositionable shelves that electrically and/or communicatively couple to a vertical power and data bus, and showing a front sensing zone including a high-density antenna matrix and an LED information strip configured to provide item-level guidance and pricing information.

FIG. 2 is a side or sectional view illustrating an example controller and transaction architecture for the cabinet, including a transaction processing module configured to establish a user session via biometric identification and/or scanning of an optical code generated by a mobile application, to reserve funds via a pre-authorization token, to maintain an offline financial ledger, to generate fiscal receipts upon capture, and to route settlement proceeds to one or more destinations on an item-by-item basis; FIG. 2 further illustrates optional network connectivity to one or more remote servers that may provide inventory manifests, pricing updates, supplier interfaces, and analytics.

FIG. 3 is a detail view of example shelf and sensing geometry showing a combination of (i) a front-zone antenna matrix, (ii) an overhead imaging sensor, and (iii) load cells and/or weighing structures arranged to detect removal, return attempts, and partial-consumption behaviors; FIG. 3 further illustrates a secure compartment that may include switchable smart glass and a dual-mode operating configuration for standard goods and restricted goods.

FIG. 4 is a high-level process diagram illustrating example inventory induction and transaction workflows, including a local induction station and/or remote upstream induction in which tracking tags are associated with product data at a supplier location and an allow-list inventory manifest is synchronized to the cabinet controller for recognition of pre-tagged goods upon arrival.

DEFINITIONS

Unless otherwise stated, terms such as “may,” “can,” “in some embodiments,” and “example” indicate optional, non-limiting features. A “module,” “engine,” or “controller” may be implemented as software, firmware, hardware, or combinations thereof. “AI” includes rule-based logic, statistical models, and machine learning models constrained by safety/policy rules. A “digital passport” refers to machine-readable processing parameters (e.g., target temperature, heating profile, ingredient metadata) associated with a product. “External market data” refers to data obtained from networked sources including databases, feeds, APIs, and/or scraped competitor prices. A “profiled guide tray/channel” refers to a dispensing tray defined by a cross-sectional geometry that cradles and guides the container body.

DETAILED DESCRIPTION

A. System Overview and Modular Deployment

In some embodiments, the autonomous retail ecosystem includes one or more cabinets configured to store goods and to permit controlled access to the goods by authorized users. A cabinet may include a chassis defining an interior volume, one or more doors or access panels, and modular shelving configured to support interchangeable modules. Modules may include, by way of example, refrigerated storage modules, frozen storage modules, dry-goods storage drawers or bins, produce hoppers, weighing stations, utensil or accessory dispensers, and thermal processing modules. In some embodiments, the chassis includes a universal power distribution interface that provides power to modules through a vertical busbar, a cable chain, a suspended loop, or another distribution structure that permits rearrangement of modules without custom wiring. In some embodiments, the ecosystem is deployable as a standalone cabinet having local compute, payment, and access control. In some embodiments, the ecosystem is deployable in a master-slave arrangement in which a master cabinet includes a primary compute and payment stack and communicates with one or more slave cabinets over a local bus, such that certain high-cost components are shared across cabinets while each cabinet maintains local sensing and actuation for its own doors and modules.

B. Edge Controller, Communications, and Offline Operation

In some embodiments, the cabinet includes an edge computing controller operatively coupled to the distributed sensor network and to one or more actuators, including electronic locks, lighting elements, motors, heaters, valves, and ventilation components. The edge controller may include one or more processors, memory, and persistent storage to buffer logs, transaction records, and sensor events. In some embodiments, the cabinet communicates with a remote server or cloud service via wired Ethernet, Wi-Fi, and/or cellular communications, such as LTE. In some embodiments, the cabinet includes an uninterruptible power supply or backup battery configured to maintain access control, logging, and safe shutdown behaviors during a power interruption. In some embodiments, the edge controller supports an offline mode in which the cabinet continues to authorize access, operate doors, execute local pricing or policy checks, and log events when an active internet connection is unavailable. When connectivity is restored, buffered records may be synchronized to a remote service for reconciliation, reporting, and analytics.

C. Identity, Authentication, and Access Control

In some embodiments, access to stored goods is controlled by an authentication workflow executed by the edge controller. Authentication may be performed using one or more factors, including biometrics (such as facial recognition, palm recognition, or fingerprint), presentation of an NFC credential, card reader input, a barcode or QR code token, and/or a mobile application credential. In some embodiments, the cabinet includes a touch display or tablet configured to present a user interface, including prompts, inventory visibility, pricing, and instructions. In

some embodiments, the cabinet includes a biometric camera positioned to capture facial imagery for verification and may include anti-spoofing checks. Upon successful authentication, the edge controller may unlock one or more doors using an electronic lock, illuminate one or more shelves using addressable LED strips, and guide the user to an authorized compartment or shelf. In some embodiments, the cabinet includes a mechanical key override to permit authorized maintenance access when electronics are unavailable.

D. Distributed Sensing for Inventory and Anti-Theft Verification

In some embodiments, the cabinet includes a distributed sensor network configured to detect item presence, identity, and removal. The sensor network may include RFID readers, NFC readers, shelf-edge or drawer-mounted antenna arrays, cameras, load cells, door status sensors, magnetic reed switches, proximity sensors, and environmental sensors such as temperature and humidity sensors. In some embodiments, each item is associated with an identifier, such as an RFID tag, barcode, or other machine-readable code. In some embodiments, shelf-edge antenna arrays are positioned to read tags corresponding to items placed at the associated shelf region, thereby allowing slot-level mapping and verification. In some embodiments, a camera provides a field of view over one or more shelves and is used to detect hands entering a protected region, to validate that authorized items are removed, and to generate alerts for anomalous behavior. In some embodiments, a door status sensor indicates open/closed state and cooperates with the lock to ensure that access events are recorded and time-bounded.

E. Produce Zone and Weighing Architecture

In some embodiments, the cabinet includes a produce station configured to dispense loose items and to verify removal by weight. In some embodiments, loose produce is stored within modular drawers or bins that provide humidity control rather than open shelving. In some embodiments, a conical hopper directs loose items toward a single central load cell. The geometry of the hopper may reduce or eliminate corner-load errors and may reduce calibration complexity relative to multi-sensor arrays. In some embodiments, the controller computes a dispensed weight for each removal event and may require that a total exit weight measured at an exit gate correspond, within a tolerance, to the sum of weights dispensed from loose-item hoppers and packaged-item scales during the session. In some embodiments, an access gate is opened only when the total measured exit weight is consistent with authorized removals.

F. Hybrid Thermal Zoning and Induction Processing

In some embodiments, the cabinet includes a hybrid thermal architecture having a chilled storage zone maintained at a target temperature range, such as about +2°C to +4°C, and a thermal processing module configured to heat selected items on demand. In some embodiments, the thermal processing module is an induction heating chamber configured as a hermetically sealable compartment. A package for an item may include a susceptor-bearing layer configured to couple to an induction coil and convert electromagnetic energy to heat at selected regions of the

package. In some embodiments, a product-specific cooking profile is associated with the item identifier, and the controller selects heating power, timing, and optional agitation based on the profile. In some embodiments, the heating module may include temperature sensing, steam sensing, or other instrumentation to confirm completion and safety.

G. Ventilation, Condensation Control, and Sanitation Handling

In some embodiments, the cabinet includes an active negative-pressure ventilation system configured to evacuate steam and odors generated during heating. In some embodiments, the ventilation system includes a duct coupled to the heating chamber, one or more fans, and a filter or condenser structure, and is controlled to draw vapor away from chilled storage regions during a heating cycle. In some embodiments, the cabinet includes an automated sanitation purge mechanism configured to divert a heated item to a waste module, which may be maintained below a threshold temperature (for example, below about -2°C), when the item is not retrieved within a configurable timeout. In some embodiments, such diversion may be logged and may trigger a transactional rule, such as a penalty or forfeiture consistent with posted policies.

H. Mechanical Dispensing and Anti-Reverse Features

In some embodiments, items are presented to users through a profiled guide tray or dispensing channel configured to guide containers along a defined path. In some embodiments, the cabinet includes one or more pushers configured to bias items toward an access point. In some embodiments, a mechanical anti-reverse system includes ratchets or one-way elements on product pushers to physically prevent re-insertion or return of a dispensed item to a storage position after retrieval. In some embodiments, the dispensing channel is configured to accept containers of varying heights or volumes that share a common footprint, thereby permitting standardized storage while accommodating product variation.

I. Dynamic Shelf Edge Presentation, Pricing, and Space-Yield Optimization

In some embodiments, the cabinet includes a dynamic shelf-edge display, such as a full-width LCD or video rail disposed along a shelf edge, configured to present item information. In some embodiments, the shelf-edge display presents advertising or brand content in an idle mode, and switches to a proximity mode to present item-specific information such as price, weight, and expiry date when a user is detected near the shelf. In some embodiments, the shelf-edge display presents warning indications when an expiry date is near. In some embodiments, the controller executes space-yield optimization routines that allocate inventory to shelves, bins, or modules based on demand predictions, expiry, and temperature constraints, and may recommend restocking or relocation actions to maximize utilization and revenue per unit volume.

Broad Applicability to Open Retail: While the embodiments described herein refer to a secure cabinet, it is understood that the 'Partitioned Multi-Tenant Control Architecture', the 'Space-Yield AI Engine', and the 'Instant Split-Payment Smart Contract' are not limited to enclosed

vending systems. These methods may be implemented in standard open-retail environments (e.g., supermarket shelving) equipped with Electronic Shelf Labels (ESL/LCD) and cameras. In such embodiments, the system allows third-party suppliers to remotely control pricing of open inventory slots and ensures that transaction funds are automatically split and routed immediately to the supplier and the retailer/landlord upon checkout, without requiring a centralized holding period.

J. Restocking, Expiry Association, and Audit Trail Generation

In some embodiments, the cabinet supports a restocking workflow executed by authorized staff. A staff member may scan an item using a handheld scanner, identify the item by tag or code, and associate item attributes such as an expiry date, lot number, or pricing tier with the item identifier in a local or remote database. In some embodiments, the system maintains an audit trail that logs stocking events, access events, item identity reads, removals, and transactions, including timestamps and actor identifiers. In some embodiments, the audit trail is cryptographically protected or otherwise hardened against tampering, and may be synchronized to a remote service for reporting.

K. Example User Transaction Workflow

In some embodiments, a user initiates a session by authenticating at the cabinet. The controller may unlock a door or compartment, illuminate the authorized region, and permit the user to remove one or more items. In some embodiments, the controller determines removed items based on tag reads, weight changes, and/or computer vision. In some embodiments, if a network connection is available, the controller may validate session policies and execute payment through a remote service. If a network connection is not available, the controller may enforce local database rules and may block the session, raise an alarm, or proceed with a locally authorized transaction subject to later reconciliation. Upon determining that an item has been removed and authorized, the controller records the event and executes payment or updates a virtual cart, and the session is closed when the door is secured and verification checks are satisfied.

CLAIMS

What is claimed is:

1. An Autonomous Retail Ecosystem comprising:

A reconfigurable master cabinet chassis having a universal power distribution interface selected from a vertical busbar, a cable chain, or a suspended loop; a modular shelving system supporting interchangeable modules; a distributed sensor network; and an edge computing controller;

Wherein the system further comprises:

- (a) A Hybrid Induction Heating Module configured as a hermetically sealable chamber to store items at a chilled temperature and perform on-demand heating using an induction coil interacting with a susceptor-based packaging layer, controlled by an RFID cooking profile;
- (b) An Active Negative-Pressure Ventilation System configured to evacuate steam and odors directly from the chamber during the heating cycle to prevent condensation in the chilled zone;
- (c) An Automated Sanitation Purge Mechanism configured to mechanically divert a heated item into a Sub-Zero Waste Module (maintained below -2°C) if not retrieved within a timeout period, thereby arresting bacterial growth and triggering a transactional penalty;
- (d) A Mechanical Anti-Reverse System comprising ratchets on product pushers that physically prevent the return of dispensed items; and
- (e) A Smart Glass Interface (PDLC) configured to transition from an opaque state to a transparent inspection state only upon biometric verification.

2. The System of Claim 1 (Produce & Weighing):

Further comprising a Secure Open-Zone Produce Station characterized by:

- (i) Conical Hopper Scales featuring a specific 45-degree geometry to gravitationally direct loose items to a single central load cell, thereby eliminating corner-load errors without requiring multi-sensor arrays;
- (ii) Active Auto-Leveling Feet comprising motorized actuators and gyroscopic sensors configured to physically level the chassis to ensure gravimetric accuracy;
- (iii) A Shrinkage Compensation Engine configured to adjust unit pricing exponentially based on real-time moisture loss; and
- (iv) A Total-Sum Verification Gateway that tracks the cumulative weight of all dispensed items (loose and packaged) and actuates an exit barrier only upon confirming a match with the total weight measured at the exit platform.

3. The System of Claim 1 (Physical AI Pricing):

Wherein the controller executes a localized Space-Yield Displacement Engine configured to physically optimize cabinet inventory allocation by adjusting unit prices based on a weighted aggregate of:

- (i) Optical and Gravimetric Sensory Decay: Real-time correlation of surface texture analysis (via cameras) and moisture weight loss (via scales);
- (ii) Spatial Opportunity Cost: A calculated metric representing the

revenue potential of the physical volume occupied by a slow-moving item versus a high-velocity alternative;

(iii) Contextual Demand Fluidity: Real-time adjustment factors derived from local time-of-day, weather conditions, and proximity-based competitor pricing data;

Thereby inducing a localized accelerated sales velocity to physically clear shelf space.

4. The System of Claim 1 (Ecosystem & Inter-Device):

Further comprising a Smart Cooler Synchronization protocol, wherein the cabinet transmits a dispensing command to a proximate water dispenser to release a precise volume of water (e.g., 350ml) corresponding to a specific SKU purchased in the cabinet, and a Smart Cutlery Dispenser that releases a specific utensil (fork/spoon/chopstick) matched to the product type.

5. The System of Claim 1 (Supplier Logic):

Further comprising a Partitioned Multi-Tenant Control Architecture, allowing distinct supplier entities to independently control pricing of their respective inventory slots via a remote application, utilizing an Instant Split-Payment Smart Contract that divides transaction revenue between the platform operator and the supplier in real-time.